

- ✓ This prob dist needs to be tested across various experiments and $d_{1+2} = d_1 + d_2$ also observed as one output
- ✓ If light = monochromatic, then E is same for all, and w , too.
(λ = same for all)

④ De-Broglie's Each particle has wavelike behavior.

$$k = \frac{p}{\hbar}$$

Young's exp ~~was~~ done with electrons by Davisson.

Q: Young's exp done with $m = 1 \text{ kg}$ $v = 1 \text{ cm/s}$

$$\lambda = \frac{2\pi}{k} = \frac{h}{p} \approx 10^{-26} \text{ cm (small)}$$

and we will also observe $d_{1+2} = d_1 + d_2$ (for macroscopic particles)

* Postulates of QM: (non-relativistic)

Take a single particle. $|\psi\rangle = \text{ket}$ $\langle\psi| = \text{bra}$

Classical Mech Quantum Mech

① State of particle at a given time defined by $x(t)$ and $p(t)$ as a point in phase space

④ State of particle is represented by vector $|\psi(t)\rangle$ in Hilbert's space.

② Every dynamical variable $u = u(x, p)$ is represented by Hermitian operators X, P .

x = position and p = momentum

and have the matrix element in the eigen basis of X

$$\langle x | X | x' \rangle = x \delta(x - x')$$

$$\langle x | P | x' \rangle = -i\hbar \delta'(x - x')$$

4. OM, $\delta = \text{dirac, delta function}$ change variable
 $\delta' = \frac{\partial \delta}{\partial x}$ and $\Omega(x, p) \equiv \omega(x \rightarrow x, p \rightarrow p)$
 and any dynamic any dynamic variable.
CM

③ If a particle is in state OM given by x, p ; measurement of ω with given $\omega(x, p)$
 $\omega(x, p) \Rightarrow$ doesn't effect position of particle.

③ If a particle is in state OM $|\psi\rangle$, measurement of a variable corresponding to operator Ω will yield one of the eigen values of Ω with prob.

$P(\omega) \propto |\langle \omega | \psi \rangle|^2$
 * State of the system will change from $|\psi\rangle$ to $|\omega\rangle$ as a result of measurement.
 $\omega = \text{eigen vector}$

④ State variables changes with time according to Hamilton's equation.

④ The state vector $|\psi(t)\rangle$ obeys the schrodinger eqn.

$$\dot{x} = \frac{\delta H}{\delta p}$$

$$\dot{p} = -\frac{\delta H}{\delta x}$$

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$$

$H(x, p) \rightarrow H(x \rightarrow x, p \rightarrow p)$
 Quantum Hamiltonian operator $\Downarrow$ classical Hamiltonian and change x, p to operators.

* Hilbert space $\equiv$ abstract vector space and contains various state vector like $|\psi(t)\rangle$

$|\psi\rangle = \text{ket} \in \text{Hilbert space}$

$$\langle \psi | a^\dagger = a | \psi \rangle$$

$|\psi\rangle$ corresponds to the probability amplitude $\psi(x)$

✓ If $|\psi\rangle, |\psi'\rangle \Rightarrow$ show the possible states of a particle.
 then $\alpha|\psi\rangle + \beta|\psi'\rangle =$ is also a possible state of the particle.
 (linear combination)

④ This ~~principle~~ is called as "Superposition Principle".
 (This is a postulate)

⑤ Fundamental diff. b/w CM and QM:
In CM: x, p are known
 Then we measure $w(x, p) =$ dynamical variable

In QM:
 ① For dynamical variables here, we construct QM operator Ω (Ω is hermitian)
 $w(x, p) \xrightarrow{\text{(replaced with)}} \Omega(x, p)$

$$H = \frac{p^2}{2m} \quad \hat{H} = \frac{\hat{p}^2}{2m}$$

② Find eigenvalues and eigen vector of Ω
 (w_i) $(|w_i\rangle)$
 and $w_i =$ real values because $|w_i\rangle$ These form a basis
 (See ~~base~~ is linearly independent vectors)

③ Expand $|\psi\rangle$ in basis of $|w_i\rangle$

$$|\psi\rangle = \sum |w_i\rangle \langle w_i | \psi \rangle$$

④ When we measure the Ω we get only one of the eigen values and

$$\text{the relative prob. of getting } \omega_i \propto |\langle \omega_i | \psi \rangle|^2$$

So, $P(\omega_i) \propto |\langle \omega_i | \psi \rangle|^2$

relative prob.

Absolute probability $P(\omega_i) = \frac{|\langle \omega_i | \psi \rangle|^2}{|\langle \psi | \psi \rangle|} = |\langle \omega_i | \psi \rangle|^2$

= 1 (if $\psi = \text{unit vector}$)

④④ If $|\psi\rangle$ happens to be one of the eigen states $|\omega_i\rangle$ then measurement will definitely give $|\omega_i\rangle$ with prob. 1.

④④ Say,

$|\omega_1\rangle, |\omega_2\rangle = \text{two eigen states of operator } \Omega$
 and $|\psi\rangle = \frac{\alpha|\omega_1\rangle + \beta|\omega_2\rangle}{\sqrt{\alpha^2 + \beta^2}}$ (from superposition)

$$\langle \psi | \psi \rangle = 1$$

Prob of getting $\omega_1 = \frac{|\alpha|^2}{\alpha^2 + \beta^2} = \left(\frac{\alpha}{\sqrt{\alpha^2 + \beta^2}} \right)^2$

$$\omega_2 = \frac{\beta^2}{\alpha^2 + \beta^2}$$

Say if another operator Λ , find eigen values λ_i and eigenvectors $|\lambda_i\rangle$

prob of getting $\lambda_i = \frac{|\langle \lambda_i | \psi \rangle|^2}{\langle \psi | \psi \rangle}$



normalized wave function
 $\langle \psi | \psi \rangle = 1$

① Consider 3-D Hilbert space

eigen vectors of $\hat{A} = |u_1\rangle, |u_2\rangle, |u_3\rangle$

$$\text{say, } |\psi\rangle = \frac{1}{2}|u_1\rangle + \frac{1}{2}|u_2\rangle + \frac{1}{\sqrt{2}}|u_3\rangle$$

prob of getting $|u_1\rangle$ from $|\psi\rangle = \frac{1}{4}$

$$= \frac{|\langle\psi|u_1\rangle|^2}{\langle\psi|\psi\rangle} = \frac{|\frac{1}{2}|^2}{1} = \frac{1}{4}$$

$$= \frac{|\langle\psi|u_2\rangle|^2}{\langle\psi|\psi\rangle} = \frac{|\frac{1}{2}|^2}{1} = \frac{1}{4}$$

② Collapse of state vector:

$$|\psi\rangle = \sum_w |w\rangle \langle w|\psi\rangle$$

$$\text{Ex: } |\psi\rangle = \frac{1}{2}|u_1\rangle + \frac{1}{2}|u_2\rangle + \frac{1}{\sqrt{2}}|u_3\rangle$$

after measuring state changes drastically from $|\psi\rangle$ to either $|u_1\rangle / |u_2\rangle / |u_3\rangle$

depending on which eigen value we are getting.

③ This is called state collapsing.

* Here we assume eigen values are discrete and distinct

** In QM,

we can have continuous eigen values, for position vector, we have continuous eigen values.

then,

$$|\psi\rangle = \int |u\rangle \langle u|\psi\rangle du$$

smooth function = wave function $\psi(u)$ in u space

If $u = \text{position}$, $\psi(u) = \text{wave fn in position space}$.

Then, probability density $P(u) \propto |\langle u | \psi \rangle|^2$

$$\text{total prob} = \int_{\text{all } u} P(u) \cdot du = 1 \Rightarrow \int |\langle u | \psi \rangle|^2 \cdot du = 1$$

$$= \int \underbrace{\langle \psi | u \rangle}_{\text{}} \cdot \underbrace{\langle u | \psi \rangle}_{\text{}} \cdot du = 1$$

$$\langle \psi | u \rangle \langle u | \psi \rangle = \langle u | \psi \rangle \langle \psi | u \rangle = \langle \psi | \psi \rangle$$

$$= \int |u\rangle \cdot \langle u| \cdot du = 1$$

* Expectation value of measurements:

$$\langle \Omega \rangle = \text{expectation} = \text{avg} = \sum_i P(u_i) \cdot u_i$$

$$= \sum_i |\langle u_i | \psi \rangle|^2 \cdot u_i$$

$$= \sum_i \langle \psi | u_i \rangle \cdot \langle u_i | \psi \rangle \cdot u_i$$

WKT, $\Omega |u_i\rangle = u_i |u_i\rangle$ (eigen vector)

$$\text{then } \langle \Omega \rangle = \sum_i \langle \psi | \underbrace{\Omega | u_i \rangle}_{= u_i | u_i \rangle} \cdot \langle u_i | \psi \rangle$$

$$\langle \psi | \Omega | \psi \rangle = \langle \Omega \rangle$$

For large no. of experiment we can talk about this $\langle \Omega \rangle$

$$0 = \langle \psi | (\Omega - \langle \Omega \rangle) (\Omega - \langle \Omega \rangle) | \psi \rangle$$

* Uncertainty:

$$\Delta \Omega = \text{standard deviation} = \text{uncertainty}$$

$$= \langle (\Omega - \langle \Omega \rangle)^2 \rangle^{1/2} = [\dots]$$

$$\Delta \Omega = \left[\langle \psi | (\Omega - \langle \Omega \rangle)^2 | \psi \rangle \right]^{1/2}$$

④ Say we want to measure one variable after another.
 Say Ω , Λ are variables
 and $|\psi\rangle$ - ensemble.

① First measure Ω : take the particles that are in state $|\omega\rangle$

② measure Λ : take the particles that are in a state $|\lambda\rangle$

If $|\lambda\rangle$ is eigen vector of Ω then i.e. if

Ω, Λ have some eigen vectors then

the particles in $|\omega\rangle$ after ① remain same and don't change after ②.

i.e. if Ω, Λ have common eigen vectors,

$$\Omega |\omega\lambda\rangle = \omega |\omega\lambda\rangle \quad \text{and} \quad \Lambda |\omega\lambda\rangle = \lambda |\omega\lambda\rangle$$

$$\Lambda |\omega\lambda\rangle = \lambda |\omega\lambda\rangle \quad \text{and} \quad \Omega |\omega\lambda\rangle = \omega |\omega\lambda\rangle$$

$$\Lambda \Omega |\omega\lambda\rangle = \omega \cdot \Lambda |\omega\lambda\rangle = \omega \lambda |\omega\lambda\rangle$$

$$\Omega \Lambda |\omega\lambda\rangle = \lambda \cdot \Omega |\omega\lambda\rangle = \lambda \omega |\omega\lambda\rangle$$

$$\text{i.e. } (\Lambda \Omega - \Omega \Lambda) |\omega\lambda\rangle = 0$$

$$\Rightarrow [\Omega, \Lambda] = 0 \quad \text{or} \quad \boxed{\Lambda \Omega - \Omega \Lambda = 0}$$

④ $[a, p] = i\hbar$ (two operators that don't commute).

$$[\langle \psi | \sim \langle \psi | a - a | \psi \rangle]$$

ie in QM,

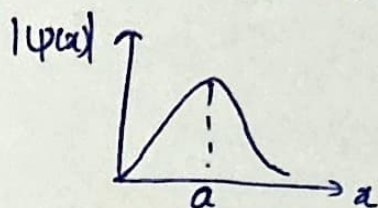
we can never have a state where we can measure p, x simultaneously $\Rightarrow$ (Heisenberg's uncertainty principle)

④ In continuous basis $|a\rangle$

$$|\psi\rangle = \int_{-\infty}^{\infty} |a\rangle \underbrace{\langle a|\psi\rangle}_{\psi(a)} da = \int_{-\infty}^{\infty} |a\rangle \cdot \psi(a) da$$

where $\psi(a)$ = wave function = $\langle a|\psi\rangle$

Ex: Take $\psi(a) = A \cdot e^{-\frac{(a-a)^2}{2\Delta^2}}$ (Gaussian)



for normalising,

$$\langle \psi|\psi \rangle = 1$$

$$1 = \int_{-\infty}^{\infty} \langle \psi|a \rangle \langle a|\psi \rangle da$$

$$1 = \int_{-\infty}^{\infty} |\langle a|\psi \rangle|^2 da = \int_{-\infty}^{\infty} (\psi(a))^2 da$$

$$1 = \int_{-\infty}^{\infty} A^2 \cdot e^{-\frac{(a-a)^2}{\Delta^2}} da = A^2 \cdot (\pi \Delta^2)^{1/2}$$

$$\boxed{A = \frac{1}{(\pi \Delta^2)^{1/4}}}$$

Normalised wave fn = $\psi(a) = \frac{1}{(\pi \Delta^2)^{1/4}} e^{-\frac{(a-a)^2}{2\Delta^2}}$

prob of finding particle b/w $a, a+da$

$$= p(a) \cdot da = |\psi(a)|^2 da = \frac{1}{(\pi \Delta^2)^{1/2}} \cdot e^{-\frac{(a-a)^2}{\Delta^2}} da$$

$$\begin{aligned}\langle a | x | \psi \rangle &= \int_{-\infty}^{\infty} \underbrace{\langle a | x | \psi \rangle}_{x \cdot \delta(x-a)} \cdot \langle a' | \psi \rangle \cdot da' \\ &= \int_{-\infty}^{\infty} x \cdot \delta(x-a) \cdot \psi(a') \cdot da' = a \cdot \psi(a)\end{aligned}$$

→ Postulate (ii)

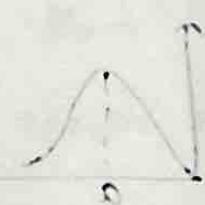
[Property of δ function]

Note:

① $\delta(x-a) = \text{dirac delta function} = \langle a | a' \rangle$

② $\delta(x-a) = 0$ for $x \neq a$

③ $\int_a^b \delta(x-a) \cdot dx' = 1$ for $a < x < b$



③ $\int \delta(x-a) \cdot f(x') \cdot dx' = f(a)$

$\delta'(x-a) = \frac{d}{dx} \delta(x-a) = -\frac{d}{dx'} \delta(x-a)$

Now,

$$\langle a \rangle = \langle \psi | x | \psi \rangle$$

$$= \int_{-\infty}^{\infty} \langle \psi | x \rangle \langle x | \psi \rangle \cdot dx$$

$$= \int_{-\infty}^{\infty} \psi^*(x) \cdot x \cdot \psi(x) \cdot dx$$

$$= \int_{-\infty}^{\infty} |\psi(x)|^2 \cdot x \cdot dx$$

For $\psi(x) = A \cdot e^{-\frac{(x-a)^2}{2\Delta^2}}$ $A = \left(\frac{1}{2\Delta^2}\right)^{1/4}$

$$\langle x \rangle = \int_{-\infty}^{\infty} x \cdot \left(\frac{1}{2\Delta^2}\right)^{1/2} \cdot e^{-\frac{(x-a)^2}{\Delta^2}} dx$$

$$\boxed{\langle x \rangle = a}$$

$$\Delta x = \text{uncertainty} = \left[\langle \psi | (x - \langle x \rangle)^2 | \psi \rangle \right]^{1/2}$$

$$= \left[\langle \psi | x^2 - 2x\langle x \rangle + \langle x \rangle^2 | \psi \rangle \right]^{1/2}$$

$$= -2\langle x \rangle \cdot \langle \psi | x | \psi \rangle = -2\langle x \rangle^2$$

$$= \left[\langle \psi | x^2 - \frac{\langle x \rangle^2}{a} | \psi \rangle \right]^{1/2} = \left[\langle \psi | x^2 | \psi \rangle - \langle \psi | a^2 | \psi \rangle \right]^{1/2}$$

$$\langle \psi | x^2 | \psi \rangle = \int_{-\infty}^{\infty} \langle \psi | x \rangle \langle x | x^2 | \psi \rangle dx$$

$$= \int_{-\infty}^{\infty} \psi^*(x) \cdot x^2 \cdot \psi(x) dx = \frac{1}{(2\Delta^2)^{1/2}} \int_{-\infty}^{\infty} x^2 \cdot e^{-\frac{(x-a)^2}{2\Delta^2}} dx$$

$$= \frac{\Delta^2}{2} + a^2$$

$$\Delta x = \left[\frac{\Delta^2}{2} + a^2 - a^2 \right]^{1/2} = \boxed{\frac{\Delta}{\sqrt{2}} = \Delta x}$$

④ We want to know the prob distrib of diff values of P .

① construct P in this basis.

② Find eigen values p and eigen vectors $|p\rangle$

③ Find $|\langle p|\psi\rangle|^2 dp$

$$\begin{aligned}\langle x|P|\psi\rangle &= \int_{-\infty}^{\infty} \langle x|P|x'\rangle \langle x'|\psi\rangle dx' \\ &= \int_{-\infty}^{\infty} -i\hbar \delta'(x-x') \psi(x') dx' \quad \text{postulate ①} \\ &= \int_{-\infty}^{\infty} -i\hbar \delta'(x-x') \cdot \psi(x') dx'\end{aligned}$$

④ Trick:

$$x = \boxed{-i\hbar \frac{d\psi(x)}{dx} = \langle x|P|\psi\rangle} \quad \delta'(x-x') = \delta(x-x') \cdot \frac{d}{dx'}$$

$$\textcircled{*} \langle x|P|\psi\rangle = -i\hbar \frac{d\psi_p(x)}{dx} \quad (\psi = p) \rightarrow \text{eqn ①}$$

$$\boxed{\psi_p(x) = \frac{-i\hbar \cdot d\psi_p(x)}{dx}}$$

ie In x basis,

$$\text{operator } p = -i\hbar \frac{d}{dx}$$

Solution for $\psi_p(x)$ is

$$\psi_p(x) = A \cdot e^{ipx}$$

$$\text{choose } A = \frac{1}{\sqrt{2\pi\hbar}}$$

$$\text{so that } \langle p|p'\rangle = \delta(p-p')$$

Eigen value eqn: $\hat{P}|\psi\rangle = p|\psi\rangle$
 In x-basis: $\langle x|\hat{P}|\psi\rangle = p\langle x|\psi\rangle$ (act on both sides)
 This led to eqn ①.

$$\langle P \rangle = \int_{-\infty}^{\infty} \psi^*(p) \cdot p \cdot \psi(p) \cdot dp$$

$$\psi(p) = \langle p|\psi \rangle$$

$$\psi(p) = \langle p|\psi \rangle = \int_{-\infty}^{\infty} \langle p|x \rangle \cdot \langle x|\psi \rangle \cdot dx$$

$$= \int_{-\infty}^{\infty} \psi_p^*(x) \cdot \psi(x) \cdot dx = \text{Fourier transform of } \psi(x)$$

$$= \int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi\hbar}} \cdot e^{-ipx/\hbar}}_{\psi_p^*(x)} \cdot \frac{e^{-\frac{(x-a)^2}{2\Delta^2}}}{(2\Delta^2)^{1/4}} \cdot dx$$

$$= \left[\left(\frac{\Delta^2}{\pi\hbar^2} \right)^{1/4} \cdot e^{-\frac{ipa}{\hbar}} \cdot e^{-\frac{p^2\Delta^2}{2\hbar^2}} \right] = \psi(p)$$

$$\langle P \rangle = \int_{-\infty}^{\infty} p \cdot \underbrace{\left(\frac{\Delta^2}{\pi\hbar^2} \right)^{1/4}}_{\psi^*(p)} \cdot \underbrace{1 \cdot e^{-\frac{p^2\Delta^2}{2\hbar^2}}}_{\psi(p)} \cdot dp$$

$\psi^*(p) \cdot \psi(p) = \text{even fun} \cdot \text{odd fun} = \text{odd fun}$

$$\langle P \rangle = 0$$

$$\Delta P = \frac{\hbar}{\sqrt{2}\Delta}$$

(check)

$$\Delta x \cdot \Delta P = \frac{\Delta}{\sqrt{2}} \cdot \frac{\hbar}{\sqrt{2}\Delta} = \frac{\hbar}{2}$$

$$\Delta x \cdot \Delta P \geq \frac{\hbar}{2} = \text{uncertainty principle}$$

lower limit

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

✓ distributions (other than Gaussian, too)

⊛ Schrödinger's Eqn:

postulate (4):

(has time evolution)

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$$

Hamiltonian operator.

Ex: Suppose, we have a 1-D Harmonic oscillator.

In CM, $H = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2$

In OM, Hamiltonian operator $H(x \rightarrow X, p \rightarrow P)$

$$= \frac{P^2}{2m} + \frac{1}{2} m \omega^2 X^2$$

$$i\hbar |\dot{\psi}\rangle = H |\psi\rangle$$

~~the eqn~~

Write the solution as $|\psi(t)\rangle = U(t) |\psi(0)\rangle$

propagator

take: $H = H(x, p)$

$$H |E\rangle = E |E\rangle \quad (\text{eigen. val eqn})$$

$$|\dot{\psi}(0)\rangle = -\frac{i}{\hbar} H |\psi(0)\rangle$$

$$|\psi(t)\rangle = \sum_{E \in \mathcal{V}} |E\rangle \underbrace{\langle E | \psi(t) \rangle}_{a_E(t)}$$

$$|\psi(t)\rangle = \sum_E a_E(t) |E\rangle$$

$$\left(i\hbar \frac{d}{dt} - H \right) |\psi(t)\rangle = \sum_E a_E(t) \cdot \left(i\hbar \frac{d}{dt} - H \right) \cdot |E\rangle$$

(schrodinger eqn) $= \sum_E a_E(t) \cdot |E\rangle \left(i\hbar \frac{d}{dt} - H \right)$

ie $0 = \sum_E |E\rangle \left(i\hbar \frac{da_E}{dt} - H a_E(t) \right)$

$|E\rangle$ = independent vector, ie every coeff = 0

$\Rightarrow i\hbar \frac{da_E}{dt} - H a_E(t) = 0$

$\Rightarrow \frac{da_E}{dt} = \frac{H}{i\hbar} a_E(t) \quad \boxed{a_E(t) = e^{-\frac{iH}{\hbar} t} a_E(0)}$

So, $\boxed{a_E(t) = a_E(0) \cdot e^{-iEt/\hbar}}$

$\langle E | \psi(t) \rangle = \langle E | \psi(0) \rangle \cdot e^{-iEt/\hbar}$

$|\psi(t)\rangle = \sum |E\rangle \langle E | \psi(t) \rangle$
 $= \sum |E\rangle \langle E | \psi(0) \rangle \cdot e^{-iEt/\hbar}$

$U(t) = \sum |E\rangle \cdot \langle E | e^{-iEt/\hbar}$

④ In schrodinger Eqn,

$i\hbar \dot{\psi} = H \psi$

we took, $|\psi(t)\rangle = U(t) \cdot |\psi(0)\rangle$
 propagator

H is explicitly time (t) independent $\rightarrow$ doubt

$|\dot{\psi}(0)\rangle = -\frac{H i}{\hbar} \cdot |\psi(0)\rangle$

We know, $H|E\rangle = E|E\rangle$ (eigen vector of H)

$$|\psi(t)\rangle = \sum |E\rangle \cdot \langle E|\psi(t)\rangle = \sum a_E(t) |E\rangle$$

$$|\psi(t)\rangle = -\sum a_E(t) \cdot |E\rangle$$

$(i\hbar \frac{d}{dt} - H)$ mul on both sides

$$(i\hbar \frac{d}{dt} - H) |\psi(t)\rangle = (i\hbar \frac{d}{dt} - H) \sum_E a_E(t) \cdot |E\rangle$$

(Schrodinger's eqn)

$$\Rightarrow \sum_E \left(i\hbar \frac{d a_E(t)}{dt} - H \cdot a_E(t) \right) |E\rangle = 0$$

linearly independent.

$$\Rightarrow a_E(t) = a_E(0) \cdot e^{-iEt/\hbar}$$

$$|\psi(t)\rangle = \sum a_E(0) \cdot e^{-iEt/\hbar} |E\rangle$$

$$= U(t) \cdot |\psi(0)\rangle$$

$$|\psi(t)\rangle = \sum_E |E\rangle \langle E|\psi(0)\rangle \cdot e^{-iEt/\hbar}$$

$$U(t) = \sum_E |E\rangle \langle E| \cdot e^{-iEt/\hbar}$$

Normal modes / stationary states.

$$|E(t)\rangle = |E\rangle \cdot e^{-iEt/\hbar}$$

prob dist of any observable Ω

$$P(\omega, t) = |\langle \omega | \psi(t) \rangle|^2$$

$$\text{if } \psi(t) = |E(t)\rangle,$$

$$\begin{aligned} \text{then } P(\omega, t) &= |\langle \omega | E(t) \rangle|^2 = |\langle \omega | E \rangle \cdot e^{-iEt/\hbar}|^2 \\ &= |\langle \omega | E \rangle|^2 \quad (\text{Independent of time}). \end{aligned}$$

* Schrodinger eqn, for

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi \quad (V=V(x))$$

Method of separation of variables

$$\psi(x, t) = X(x) \cdot \phi(t)$$

$$\frac{i\hbar \frac{\partial \phi}{\partial t} \cdot X}{X\phi} = \frac{-\frac{\hbar^2}{2m} \frac{d^2 X}{dx^2} \cdot \phi + VX\phi}{X\phi}$$

$$i\hbar \frac{1}{\phi} \frac{d\phi}{dt} = \frac{-\hbar^2}{2m} \cdot \frac{1}{X} \cdot \frac{d^2 X}{dx^2} + V = E \text{ (constant)}$$

$$\textcircled{i} \quad i\hbar \cdot \frac{1}{\phi} \frac{d\phi}{dt} = E$$

$$\boxed{\phi(t) = e^{-iEt/\hbar}}$$

$$\textcircled{ii} \quad \frac{-\hbar^2}{2m} \cdot \frac{1}{X} \cdot \frac{d^2 X}{dx^2} = E - V$$

$$\frac{-\hbar^2}{2m} \cdot \frac{d^2 X}{dx^2} + VX = EX$$

↓
solution depends on V

(time independent schrodinger eqn)

separation const $E = \text{Energy value}$

(Eigen value)

④ Solutions where x and t dependence can be separated are called stationary states.

Because, $\psi(x, t) = \chi(x) \cdot e^{-iEt/\hbar}$

$$|\psi(x, t)|^2 = \chi^*(x) \cdot \chi(x) \Rightarrow \text{Independent of time.}$$

① stationary states are solns of

$$\langle \hat{H} \rangle = \int \psi^* \hat{H} \psi dx = \int \chi^* \hat{H} \chi dx$$

Independent of t

② Stationary states are states with definite energy.

$$H(x, p) = \frac{p^2}{2m} + V(x) \quad (\text{classical hamiltonian})$$

$$\hat{H} = \frac{p^2}{2m} + V(x) \quad (\text{QM})$$

Time independent schrodinger eqn will be

$$\hat{H}\psi = E\psi \Rightarrow \hat{H}\chi = E\chi$$

and $\psi(x, t) = \chi(x) \cdot e^{-iEt/\hbar}$

$$\langle \hat{H} \rangle = \int \psi^* \hat{H} \psi \cdot dx$$

$$= \int \chi^* \cdot (e^{-iEt/\hbar})^* \cdot \hat{H} \cdot \chi \cdot (e^{-iEt/\hbar}) \cdot dx$$

$$= E \int |\chi|^2 dx = E$$

ie $\langle H \rangle = E \Rightarrow$ Energy eigen value.

$$\textcircled{*} \hat{H}^2 \psi = \hat{H} (\hat{H} \psi) = \hat{H} (E\psi) = E \cdot (E\psi) = E^2 \psi$$

$$\textcircled{*} \langle \hat{H}^2 \rangle = \int \psi^* \hat{H}^2 \psi dx$$

$$\Delta H \sqrt{\langle H^2 \rangle - \langle H \rangle^2} = 0$$

$\therefore$ can measure H (energy) for an eigen state with zero uncertainty because the state that will be measured is stationary state of H .

$\textcircled{3}$ General solution is a linear combination of separable soln.

$$\psi_1(x, t) = X_1(x) \cdot e^{-iE_1 t / \hbar}$$

$$\psi_2(x, t) = X_2(x) \cdot e^{-iE_2 t / \hbar}$$

General solution,

$$\psi(x, t) = \sum_{n=1}^{\infty} C_n \cdot X_n(x) \cdot e^{-iE_n t / \hbar}$$

$\textcircled{1}$ To solve time dependent schrodinger eqn;

Steps:

(i) Solve time independent schrodinger eqn.

General soln:

$$\psi(x, 0) = \sum_{n=1}^{\infty} C_n \cdot X_n(x)$$

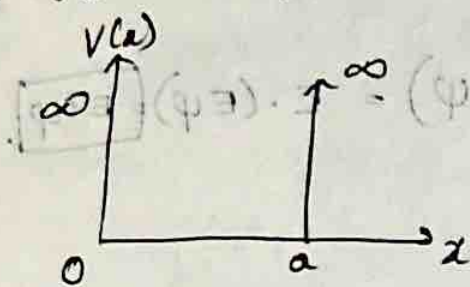
Find all $X_n(x)$.

(ii) Then write $\psi(x, t)$

$$\psi(x, t) = \sum_{n=1}^{\infty} C_n \cdot X_n(x) \cdot e^{-iE_n t / \hbar}$$

$$= \sum_{n=1}^{\infty} \psi_n(x, t)$$

Q: The ∞ -~~square~~ potential well: $E < V$



$$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{elsewhere} \end{cases}$$

$$\text{So, } \psi(x) = \begin{cases} \psi & 0 < x < a \\ 0 & \text{elsewhere} \end{cases}$$

Inside the well,

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi$$

$$\frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2} \cdot \psi = -k^2\psi$$

$E > 0$ always
why?

$$k = \frac{\sqrt{2mE}}{\hbar}$$

$$\psi = a \cdot \sin(kx + b)$$

(or)

$$\psi = A \sin ka + B \cos ka$$

A, B determined by the boundary condns.

~~find~~

$\psi, \frac{d\psi}{dx}$ continuous at $x=a, x=0$
(at the boundary)

$$\psi(0) = \psi(a) = 0$$

$$B=0 \text{ and}$$

$$A \sin ka = 0$$

$$\sin ka = 0$$

$$ka = n\pi$$

$$n = 0, \pm 1, \pm 2, \dots$$

$$n=0 \Rightarrow \psi=0 \text{ (not interested)}$$

$$k_n = \frac{n\pi}{a}$$

$$\frac{2mE_n}{\hbar^2} = \frac{n^2\pi^2}{a^2}$$

$$E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}$$

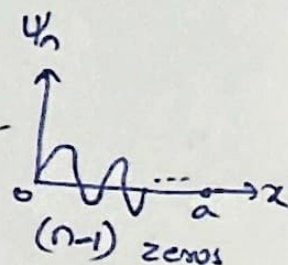
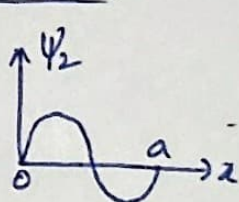
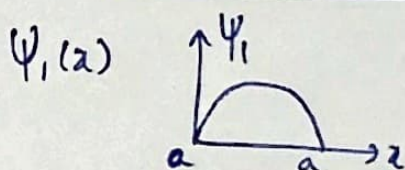
$$\left(\hbar = \frac{h}{2\pi}\right)$$

For normalising ψ

$$\int_0^a |\psi|^2 \cdot \sin^2 kx \cdot dx = 1 \quad \text{at } \frac{1}{\pi} \int_0^a \sin^2 kx \cdot dx = \frac{1}{\pi} \int_0^a \frac{1 - \cos 2kx}{2} \cdot dx$$

$$|\psi|^2 \cdot \frac{a}{2} = 1$$

$$\Rightarrow \boxed{\psi_n(x) = \sqrt{\frac{2}{a}} \cdot \sin\left(\frac{n\pi x}{a}\right)}$$



In CM, E can be 0

but in QM E can't be 0, but can have min energy.

$n=1 \Rightarrow$ lowest energy (ground state).

Notes: ① $\psi_n(x)$ = even / odd

② As one goes up in energy, no. of nodes increase by the

③ These are mutually orthogonal.

$$\int \psi_m^*(x) \cdot \psi_n(x) dx = 0 \quad (m \neq n)$$

④ These are complete set of states.

Any other fn can be expressed as linear combo of these fns.

$$f(x) = \sum_{n=0}^{\infty} C_n \cdot \psi_n(x) = \sqrt{\frac{2}{a}} \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{a}\right)$$